MILESTONE PROJECT

LIFESTYLE AND HEALTH ANALYSIS

INTRODUCTION

This project is based on a Health and Nutritional Awareness Survey dataset collected from Kaggle. The dataset is large and contains multiple lifestyle-related variables such as diet type, food consumption, physical activity, sleep duration, BMI value, gender, age, weight, health issues, etc.

The main purpose of this project is to analyze how lifestyle factors like food habits, physical activity, and sleep impact on individual’s health, especially BMI category and overall health condition.

Microsoft Excel is used as the primary tool for data cleaning, transformation, analysis, and visualization.

AIM

To analyze the relationship between lifestyle habits and health outcomes. To classify individuals into healthy and unhealthy categories. To understand how factors like food consumption, sleep duration, and physical activity affect BMI. To build an interactive dashboard for easy interpretation and decision making.

SCOPE

. Covers BMI analysis across age and gender groups.

. Studies the impact of:

Physical activity, Sleep duration, fast food, oily food and sugar consumption.

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. Provides visual insights using charts and slicers.

. Useful for healthy awareness, lifestyle improvement, basic health monitoring.

CHALLENGES FACED

. Large dataset with irrelevant columns,

Presence of duplicate records, empty cells, unnecessary brackets, commas and spaces.

. Macro security issues.

. Difficulty in linking formulas directly to the dashboard.

TOOLS USED

. Microsoft Excel

. Power Query

. Pivot Tables & Pivot Charts

. Goal Seek (What-if-Analysis)

. Macro

. Excel Formulas

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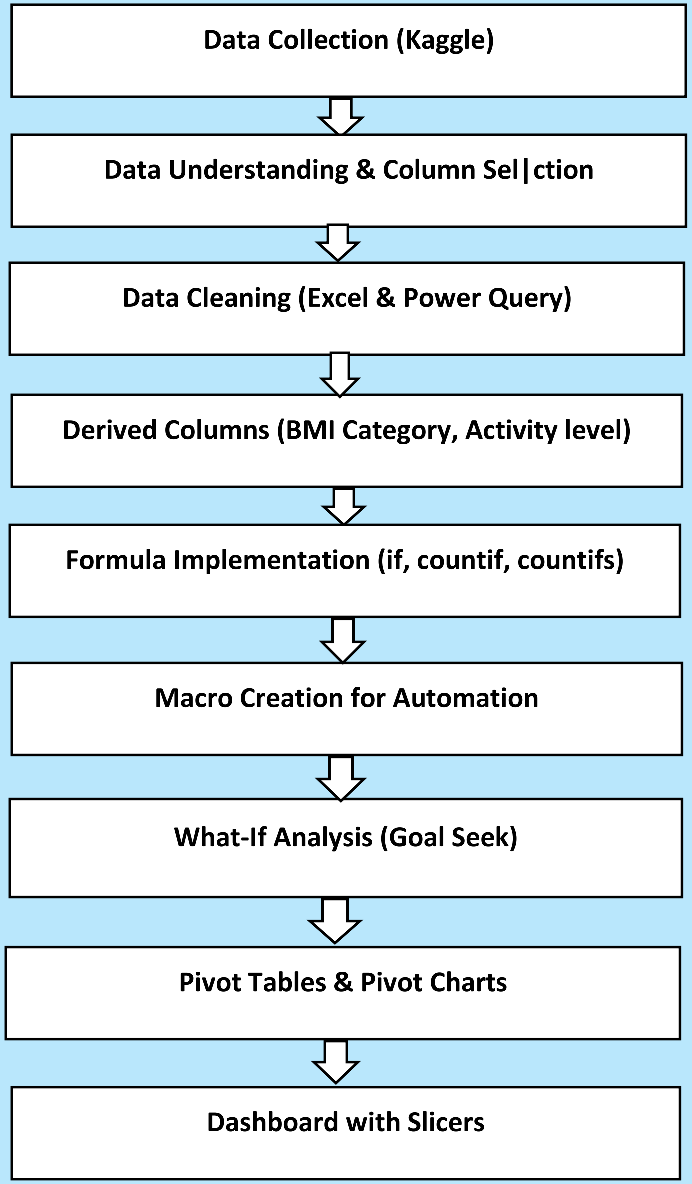
WORKFLOW DIAGRAM



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DATA CLEANING

Removed irrelevant columns . Deleted duplicate records. Handle empty spaces and blank cells. Removed unwanted brackets, commas, and symbols. Standardized and fixed column names. Ensured numerical columns were properly formatted.

Power Query was extensively used to ensured clean and structured data.

DERIVED METRICS

Using Power Query and Excel formulas, the following columns were derived:

.BMI Category (Underweight, Normal, overweight, Obese)

.Activity Level (Low, High, Moderate)

Functions Used: IF, ELSE, COUNTIF

These derived metrics helped in better segmentation and analysis.

WHAT-IF ANALYSIS

Goal Seek analysis was used to analyze BMI behaviour:

When BMI value changes, corresponding height and weight variations were identified.

Helped in understanding how small changes in physical parameters affect BMI.

AUTOMATION USING MACRO

Created a small VBA sub procedure. Added a button for one-click automation.

Automated:

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. Power Query & pivot Table refresh

This improved efficiency and reduced manual effort.

PIVOT TABLE & PIVOT CHARTS

Pivot tables were created for the following comparisons:

. Age vs Average BMI

. Gender vs BMI Category

. Physical Activity Frequency vs Health issues

. Sleep Duration vs Health Issues

. Fast Food Consumption vs BMI Category

. Oily Food Consumption vs BMI Category

. Sugar Consumption vs BMI Category

. Health Check-Up vs Health issues

CHARTS USED:

Column Chart

Line Chart

Bar Chart

Stacked Column Chart

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These visuals made the analysis easy to understand and interpret.

INSIGHTS

. Around 53% of the population is healthy, calculated using Excel formulas.

. High physical activity is strongly associated with normal BMI.

. poor sleep duration shows a higher relation with health issues.

.Increased fast food, oily food, and sugar consumption leads to overweight and obesity.

. Lifestyle choices have a direct and measurable impact on health.

CONCLUSION & RECOMMENDATIONS

The project clearly shows that lifestyle factors such as diet, physical activity and sleep duration play a major role in determining health status. Using Excel-based analysis, meaningful health insights were successfully derived.

Recommendations:

. Encourage regular physical activity.

. Maintain proper sleep duration.

. Reduce fast food, oily food and sugar intake.

. Use dashboards like this for health awareness and preventive care.

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